Total No. of Questions :10]

SEAT No. :

P3245

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B.E. (Information Technology)

MACHINE LEARNING

(2012 Course) (End Semester) (Semester-I)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Neat diagrams must be drawn wherever necessary.***
2) ***Figures to the right side indicate full marks.***
3) ***Use of Calculator is allowed.***
4) ***Assume suitable data if necessary.***

Q1) a) Why sometimes algorithms fail to produce correct results? Explain it with help of examples. **[4]**

b) Define Accuracy, True positive rate, True negative rate for binary classification tasks. **[6]**

OR

Q2) a) Draw and describe how machine learning is used to address a given task. **[6]**

b) What is a contingency table/ Confusion Matrix? What is the use of it? **[4]**

Q3) a) Define and explain model for classification. Also explain the same with respect Binary classification. **[4]**

b) Show that how the values of θ_0 and θ_1 are estimated in a linear regression model of $Y_i' = \theta_0 + \theta 1.X_i$ **[6]**

OR

Q4) a) What is a multivariate linear regression? **[4]**

b) What is overfitting ? Specify the reasons for overfitting. **[6]**

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Q5) a) List and explain various distance fuctions. **[8]**

b) Consider the following dataset: **[10]**

X	Y
–1	0.0319
0	0.8692
1	1.9566
2	3.0343

We want to learn a function f(x) = ax+b which is parametrized by a = 1 and b= 1, Considering squared error as the loss fuction, calculate the loss function.

OR

Q6) a) Define and explain single linkage, complete linkage and average linkage with neat diagram. **[8]**

b) Consider the following dataset: **[10]**

X_1	X_2	Y
2	1	4
6	3	2
2	5	2
6	7	3
10	7	3
4	4	2
7	6	3

Model this function using the K- nearest neighbor regression. What will be the value of Y for the instance $(X_1,X_2)=(3,5)$ and K=3

Q7) a) Write a note on Compression based model. **[6]**

b) A patient takes a lab test of Cancer and the result is positive. The test returns a correct positive result only in 98% of the cases, in which the disease is actually present, and a correct negative result in only 97% of the cases in which the disease is not present. Further, 0.008 of the entire population have this cancer. What is the probability that the patient has Cancer? **[10]**

OR

Q8) a) What is a normal distribution? Describe properties of normal distribution. **[8]**

b) What is the Naïve Bayes assumption. Define and explain Bayes Optimal Classifier. **[8]**

Q9) a) What is an Ensemble Model. Explain Bagging in detail. **[8]**

b) What is multi-task learning ? Explain various applications of it. **[8]**

OR

Q10) a) Explain Sequence Prediction model. What are the different measure to decide the goodness of the sequence prediction model. **[8]**

b) How does On-line learning differ from batch learning algorithms. **[8]**

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